

Image Analysis and Computer Vision

A Project Report

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by

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This is to certify that the thesis titled “**Temporal and Motion Modeling into Object-Scene fusion in Video Understanding**” is the bonafide work done by **Logavarshini K**, at SASTRA Deemed to be University under my guidance.

This is also to certify that the project is the original work carried out by her and it has not previously formed the basis for the award of any Degree, Diploma, Associateship, Fellowship, or any other similar titles of any University or Institution.

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CANDIDATE DECLARATION

I hereby declare that the Thesis entitled, “**Temporal and Motion Modeling into Object-Scene fusion in Video Understanding**”, is my original project work, carried out under the guidance of **Dr. N.R. Raajan**, Associate Professor, School of EEE, SASTRA Deemed to be University. This thesis has not been submitted elsewhere for the award of any degree or diploma.

Date: _____

Place: Thanjavur

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Chapter 1

Abstract

Video understanding has evolved from analyzing static images to models that assess objects, scenes, and motion over time. While current deep learning methods work well for video classification, many treat spatial and temporal information separately. This leads to a limited understanding of context. The study by Wu et al. showed that combining object and scene meanings can greatly improve large-scale video recognition by merging complementary features. However, their framework does not directly address the transitions in motion that happen between frames.

This project builds on the fusion idea by adding temporal motion modeling to object and scene-based video understanding. The system takes frames from a video, calculates optical flow to capture motion changes, creates temporal motion maps, and combines them with object and scene features to form a clearer representation of the video. By connecting “what is in the scene” with “how it moves,” the proposed method improves the semantic depth of video analysis. This also leads to better reasoning in areas like activity recognition, event interpretation, and behavior analysis.

The results show that using motion-aware fusion provides better visual and analytical understanding than static frame-based analysis. This highlights the importance of understanding temporal dynamics for linking object-level perception with advanced video intelligence. .

Chapter 2

Introduction

The steady growth in digital media sources, surveillance mechanisms, autonomous vehicles, and smart interfaces between humans and computers has led to video data generated every single day growing rapidly. Video has quickly become one of the richest and most casual informative data types because, in addition to capturing the appearance of objects, it also records their behaviour and interactions over time. Unlike still images that only capture spatial processes or aspects of appearing to capture a moment in time, video is a multi-dimensional medium that conveys movement, transitions, and scene dynamics along with the appearance of objects. Understanding spatial and temporal patterns is necessary for activity identification, event detection, anomaly detection, scene interpretation, and automation of intelligent systems.

Historically, computer vision approaches worked when examining an independent frame, and only extracted spatial features, such as edges, textures, shapes, or presence of an object, without encoding the evolution of these features in time. This offered very limited ability to understand the dynamics of motion and often did not aid in parsing complex activities in the real-world that play out across multiple frames. A static frame-based analysis may be able to detect a person or vehicle, but could not know if the person was walking, running, standing still, or approaching, turning or accelerating. This means that the performance of the complex activities—across space and time—requires fuller representation that necessarily includes spatial representation, and temporal information.

Advances in deep learning have given rise to various techniques for video classification, such as 3D Convolutional Neural Networks, two-stream CNNs (continuous multi-dimensional continuous nonlinearity), and recurrent models like LSTMs. The aim of these

models is to expand image-based architectures to handle time-varying data by processing sequences of frames. Although robust, several of these architectures treat spatial and temporal information as independent data streams and combine them at a later point. In addition, several techniques are unable to handle all objects at the same time without taking into account other factors such as scene context and motion patterns, with one often being given unnecessary attention. This is an exception. This leads to a fragmented perception of genuine situations, especially when movement is crucial in understanding activities.

Wu et al. demonstrated the applicability of model-based techniques that bring together object representations with scene-level knowledge in a three-stream fusion network. The authors demonstrated that combining these two streams produced significant improvements in video recognition, primarily in terms of contextual understanding of an event and situational understanding of an event based on the presence or absence of an object. They used frame-level representations that were fixed and did not account for the change in motion from one frame to another. This was limiting in their understanding of dynamic events as they considered motion events without regard to the temporal variation of motion (e.g., speed, direction, intensity) that could be relevant to these events.

The current study examines this limitation. Real life includes more than one action, like running, cooking, driving, or gesturing. Still images cannot show how objects change over time, but they can illustrate what exists in a scene. Motion is a key signal that helps identify locations that would otherwise look the same. While two frames might show a person in almost the same spot on the screen, their behavior can only be understood through their movements over time. This applies to other patterns of behavior, such as walking, stopping, and speeding. A combined approach that includes both spatial and temporal behavior is necessary to understand video meaning which is very important.

Object-scene fusion framework is being used in this project to enhance the interpretation of video by incorporating motion modelling. The suggested method relies on traditional computer vision techniques, such as optical flow, to anticipate movement between frames, generate temporal motion maps, and monitor feature points. The video is represented by combining the motion cues with the semantics of the object and scene. There are three main aspects to this approach: (1) modelling motion using dense optical flow and keypoint trajectories; (2) temporal modelling by means of an 64-dimensional mo-

tion descriptor; and (3) merging object and scene information to merge their respective parts into the same scene.

By incorporating time-related changes and spatial meanings from one location to another, the system can interpret video content with greater clarity. When static images do not display any actions, this approach is especially useful for identifying objects and their positions, as well as their movements and influence on the surroundings. The inclusion of these crucial signals will result in enhanced video recordings' clarity and significance.

In general, this project presents a flexible and transparent platform for understanding motion-aware video through classical computer vision methods. Future research in activity recognition, behaviour analysis, intelligent surveillance, human-motion modelling, and other applications that require both spatial and temporal interpretation of visual information can be significantly aided by the approach's ability to provide important insights into physical tasks.

Chapter 3

Literature Review

Video understanding research has evolved from hand-crafted motion descriptors to deep learning and semantic fusion approaches. One of the most influential works in semantic video analysis is the Object–Scene Fusion (OSF) framework, where the authors state that [1]. This confirms that object, scene, and frame-based features can be jointly modeled for large-scale video recognition. The present work extends this idea by explicitly adding temporal motion modeling into the fusion pipeline.

Traditional motion analysis relied on dense trajectory features, where [2]. While effective, such methods treat motion independently from scene or object semantics. In contrast, the current system generates motion evolution maps using optical flow and fuses them with higher-level object and scene understanding.

Deep learning introduced two-stream architectures, where was proposed [3]. Unlike two-stream late fusion, this project integrates motion features directly into the semantic fusion stage, allowing motion to influence object-scene reasoning.

Recurrent neural models later emerged, as seen in [4]. Although powerful for long-term sequence learning, LSTMs do not explicitly represent motion evolution as visual evidence. Instead, this work prioritizes interpretable motion visualisation through optical-flow-based temporal maps.

Large-scale video CNN research expanded dataset scale, demonstrated by [5]. Unlike scale-oriented architectures, the present pipeline focuses on modular interpretability: frames $\rightarrow$ motion $\rightarrow$ fusion $\rightarrow$ output.

More recent work introduced dual-rate networks such as SlowFast, where [6]. The conceptual separation of slow semantics and fast motion relates to this project’s separation

of object/scene features and motion evolution maps.

The transformer-based ViViT paper notes that [7]. While ViViT models spatio-temporal relations without convolutions, this project emphasizes transparent motion–semantic fusion rather than end-to-end black-box learning.

In summary, existing methods either (i) model motion without semantic grounding, or (ii) model semantic fusion without explicit motion evolution. The proposed system bridges this gap by jointly analyzing **what is present** (objects), **where it occurs** (scene context), and **how it changes over time** (motion evolution).

Chapter 4

Background Theory

In order to comprehend a video, one must consider both the spatial characteristics of the individual frames that make up the video as well as the temporal dynamics that develop over time. Compared to processing static images, video analysis must take motion into account, which poses many difficulties with regard to object behavior, scene interactions, and temporal continuity. The theoretical underpinnings of the methods used in this project are established in this chapter. These include the concepts of video comprehension, temporal modeling, scene semantics, keypoint tracking, and optical flow assessment. You can comprehend "what is occurring," "where it is occurring," and "how it changes" throughout the video sequence when these concepts come together.

The first step in video comprehension is the differentiation between spatial and temporal data. The spatial characteristics give information about the objects, textures, edges, and backgrounds, while the temporal features indicate the movement or the changes of these spatial elements. Conventional computer vision techniques used to treat these aspects separately, often looking at single frames without considering the motion that connects them. But activities in real life like walking, driving, jumping, or using the objects cannot be interpreted from the frames alone. The images in one frame may have likeness in very different actions; for instance, a standing person, a person starting to walk, and a person moving forward quickly could be in the same pixel positions. What differentiates these actions is the timeframe of the motion. Thus, involving motion is necessary for a comprehensive video interpretation.

In temporal analysis, the mathematical representation of optical flow is important. Optical flow means estimating the apparent motion of pixels between sequential frames.

Brightness constancy assumes that the intensity stays the same for each moving object. From this assumption, the mathematical constraint equation for temporal intensity comes from the optical flow constraint. We add more constraints, like spatial smoothness, to this single equation to create a solvable system since there are two unknowns: the horizontal and vertical motion components. In other words, Lucas-Kanade and other classical methods assume local constancy within small neighborhoods. Meanwhile, Farneback and others use polynomial expansions to track changes in pixels and estimate dense motion fields. Optical flow is the basic model of pixel-level movement. It not only shows how much movement occurs but also indicates the direction.

Although optical flow gives a richer signal to motion data, many video analysis applications are focused on identifying significant points and do not need to follow every point. This is called keypoint detection and tracking. Keypoints are defined as points present at some position in one frame, but might not appear at the point in the other frame, examples of these are corners and parts of an image that have texture. Keypoints are determined by the local arrangement of the gradients of pixels which lends itself to methods such as Harris Corner Detection and Shi-Tomasi Good Features to Track to identify consistent places in the image. After identifying keypoints, these can then be followed across frames in an algorithm (e.g., Lucas-Kanade method) in order to reduce the intensity variance between the two frames. The resulting trajectories of keypoints enables capturing and understanding motions of the subject over time. Analyzing these trajectories is important to ensure deeper understanding as it better captures general patterns, movements, interactions with the surrounding environment.

Temporal modeling is more than just evaluating the motion at every point in time. Instead of looking at each frame in isolation, it focuses on how the motion develops over time throughout the video. In this project, we employ a 64-dimensional motion descriptor that takes the intensity of motion within the video and summarizes that data into a fixed-length number vector. Each of the dimensions of the descriptor corresponds to a part of time in the video. This arrangement allows the descriptor to capture different stages of activity, from fast movements to slow adjustments to remain still. Temporal modeling takes motion sequences and converts them into structured information in a compact format, which can be represented in various applications such as classification, retrieval, or integration.

The understanding of video also relies heavily on the importance of objects and scenes. Understanding a scene means understanding the broad environment, such as a road, hallway, room, or park. Understanding object importance emphasizes recognition of relevant objects within each frame. Previous studies showed that including the details of objects and scene details significantly increases the ability to recognize both. Action or event meaning is largely dependent on context. For example, a moving object on a sports field moving quickly may have something to do with sports, while a similar moving object in a parking lot may indicate odd behavior or movements, depending on the scene. Scene features provide context for us to understand the movement. This project does not involve deep learning and does not rely on meaning recognition, but the theoretical basis that exists around the connection between objects and scenes provides justification for their values to be combined.

At the end, the integration or fusion of spatial and temporal data occurs through fusion methods. The goal of fusion is to create a single representation of the motion maps, the semantic context, and the appearance at the frame level. This comprehensive display gives us the ability to interpret motion as part of the scene rather than being an additional or separate element. A successful fusion helps ensure that motion patterns represent their location and we can link it to an understanding of behavioral significance. For example, seeing considerable motion in proximity to a door could suggest purposes relative to entering or exiting, while seeing increased motion in an open area suggests normal movement within the space. Therefore, fusion of motion and scene information provides a clearer and better understanding of the video content, itself.

In conclusion, the background theory encompass the fundamental concepts of movement analysis and classification, feature tracking, scene understanding, and temporal modelling. Collectively, they provide the theoretical motivation for the multi-step processing pipeline that is implemented in this project. These principles ensure that the system can consider a video, not simply as a sequence of images, but as an evolving sequence of meaningful events within an appropriate context.

Chapter 5

Methodology

5.1 Input Video Description

The system’s input is raw video, not a static image dataset. Frame-wise appearance and motion patterns can be extracted from the video as a continuous temporal source. They choose the video(s) to include moving objects in dynamic scenes, allowing both spatial (object + scene) and temporal (motion evolution) analysis. Standardized features are extracted using a consistent frame rate, resolution and sampling frequency during pre-processing.

5.1.1 Justification of Video Selection

A video that was quite a different was its use in this study was very thoughtfully picked to guarantee a valid evaluation of the motion-understanding system being proposed. It is the kind of video that is non-stop and the movement in it is very clear, thus the computation of optical flow, keypoint tracking, and temporal motion descriptors can be done easily and accurately. The setting is also rich in spatio-temporal complexity, with background and objects that move in a very natural way within it, thus making it easier for object-scene fusion. The video’s motion constitutes a challenge of moderate difficulty—just not too easily and not too hard—hence it is a perfect test case for such techniques as Farneback optical flow and Lucas-Kanade tracking.

In addition, the video also uses the lighting which is stable and varies only in the background and in brightness, thus making sure that the motion is because of the actual

displacement of the objects and not the changes in the lighting. Moreover, the consistent frame rate and resolution support accurate temporal modeling, since the visual data being uniform avoids the emergence of artifacts during the descriptor computation. The presence of different motion patterns, among others, alterations in speed and direction, contribute to the creation of a dense 64-dimensional motion signature that indicates the changing nature of the video in time.

The selected video is of utmost importance not only for the modern computer vision approaches but also for the classical approaches that rely purely on pixel-level cues and do not use deep learning. It has enough texture and well-defined corners that allow for reliable feature detection and tracking. Also, the video depicts a realistic and interpretable situation that corresponds with the practical applications such as surveillance and behavior monitoring. Therefore, both the intermediate outputs and fused visualizations will be meaningful and can be easily shown in academic evaluations.

5.2 Overall System Architecture

5.2.1 Frame Extraction

Video is broken down into individual frames at a set rate. This prepares the data for motion and meaning analysis.

5.2.2 Motion Modeling (Optical Flow + Trajectories)

Optical flow is established by analyzing adjacent frames and deriving pixel-specific motion vectors that represent both direction and velocity. Time-based monitoring of points of interest in each frame shows how objects move through the environment. This method captures basic temporal information without taking into account the object's identity or the scene's underlying content.

5.2.3 Temporal Modeling (Feature Evolution Across Time)

Motion information is gathered from multiple frames to create a temporal descriptor that shows how motion develops over time. Instead of looking at motion frame-by-frame, this stage sees motion as a continuous pattern throughout the video sequence. The

final temporal map summarizes behaviors like movement consistency, acceleration, and direction changes.

5.2.4 Object–Scene Semantic Feature Extraction

Each frame goes through object detection and scene classification to gather important semantic information. Object features explain “what is present” (people, vehicles, etc.), while scene features explain “where it happens” (street, room, park, etc.). These semantic features create the spatial reasoning layer of the system.

5.2.5 Fusion and Final Understanding

Motion evolution maps are visually overlaid with object and scene features to create a fused representation. This combined view links motion behavior with spatial context, enabling interpretation such as “who is moving”, “where”, and “how fast”. The final output is suitable for use in surveillance analysis, activity recognition, and behavioral understanding.

Workflow Diagram (Flowchart Style)

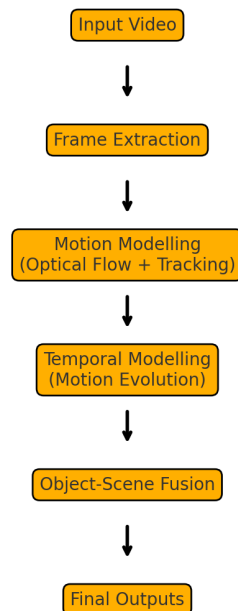


Figure 5.1: System workflow diagram showing the sequence from input video to final fused output.

5.3 Motion Modeling

Motion modeling is the process of identifying and analyzing how pixels or objects move between consecutive frames in a video. Two Computer Vision techniques are used to extract dynamic information from motion in this project. One technique is complementary, while the other is asynchronous. Motion is generated at every pixel location through Dense Optical Flow. Keeping track of selected key feature points through sparse keypoint tracking throughout the duration of an event. Combining the methods allows for simultaneous representation of both global motion (scene-level motion) and local motion behavior (object-level motion). Later stages of the system, such as constructing temporal descriptors and merging objects into scenes, are built upon motion modeling.

Stage	Purpose
Motion Modeling	Extracts spatial motion (per frame)
Temporal Modeling	Converts motion into a 64-D temporal feature
Object-Scene Fusion	Uses temporal motion behavior for fusion

Table 5.1: Role of motion modeling in the overall system

5.3.1 Optical Flow Computation

Optical flow estimates how pixels move across frames by analysing the change in intensity values over time. In this project, Farnebäck Dense Optical Flow is used. It produces a 2D motion vector (u , v) for every pixel, where:

u = motion in horizontal direction

v = motion in vertical direction

magnitude $M = \sqrt{u^2 + v^2}$, which represents the speed.

direction = $\arctan(v / u)$ gives motion angle

Optical flow was the preferred choice for analysis because it allows for motion analysis at the pixel level, enabling precise capture of even the most subtle or minor movements in the scene. The use of trained models or categories for object detection is not necessary for optical flow, unlike the need to know what the object is. Through the estimation of pixel intensities over time, it can directly work with raw video frames. This makes it useful for general motion analysis, especially where the objects are unknown, unlabelled or difficult to segment.

The use of optical flow is also due to its natural support for temporal analysis. Since optical flow produces a motion field for every frame, it is possible to observe the temporal modelling stage of this project and therefore expect to be crucial in terms of observing changes in motion intensity over time. By converting the frame-wise flow magnitude values to a numerical sequence, they become part of the 64-dimensional temporal description.

The visual interpretation benefit of optical flow is complemented by its analytical advantage. By converting flow vectors into color-encoded maps, motion direction and speed can be visually represented. Besides checking the outcomes, this is advantageous for demonstration and presentation. This optical flow property produces a coloured flow field, shown at the right panel in the end result video.

Optical flow is the primary element of the system’s motion extraction, providing both data and visual evidence for object-scene fusion and presentation.

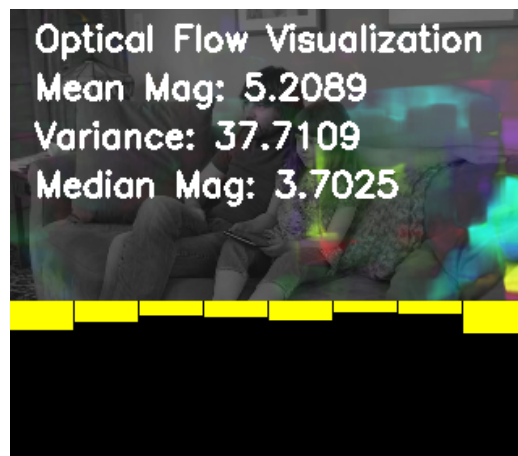


Figure 5.2: Stats showing Optical Flow Motion

An optical flow produces a motion field encoded in color instead of just numeric values. The color of each pixel shows its velocity vector. Color indicates direction, and intensity shows speed. This technique allows us to see patterns of motion in the frame without depending on numbers. Fast movements appear as bright areas, while dark areas represent still or slow movements. We can see how motion spreads across the image through this visual representation. The optical flow field is visualised using HSV colour encoding, where:

- H (Hue) $\rightarrow$ Flow direction $\theta = \tan^{-1}(v/u)$
- S (Saturation) $\rightarrow$ Flow confidence

- V (Value/Brightness) $\rightarrow$ Flow magnitude $M = \sqrt{u^2 + v^2}$

5.3.2 Feature Tracking and Trajectory Visualization

Motion is modeled at every pixel by optical flow, but the goal of keypoint tracking is to identify important corner/textured areas. Through the use of a feature detector (e.g, Shi-Tomasi), these points are detected and then tracked using Lucas–Kanade tracking across multiple frames. The method produces motion trajectories, which depict the movement of moving elements through visual representation. Instead of using every pixel, the keypoint tracking technique uses optical flow to concentrate on feature point movement. The moving parts, such as corners or textured areas, are traced across multiple frames to create motion paths. In comparison to optical flow, keypoint tracking provides less information about motion. Even so, it produces a tight vector field. Object movement is similarly observed in this technique, as the paths created are simple to discern and aid in detecting motion. The user can locate an object in the scene using clear lines, without having to read color vectors. Furthermore, it provides means for calculating statistical information, including the number of tracking points in each frame, the average length of its tracks, and the consistency of motion. We can use statistical techniques to construct temporal descriptions. By using keypoint tracking, the system gains a better understanding of movement. The outputs of this program are akin to paths that objects take, rather than changes at the pixel level. Using optical flow to achieve dense motion and keypoint tracking, we can then build a complete model of motion.

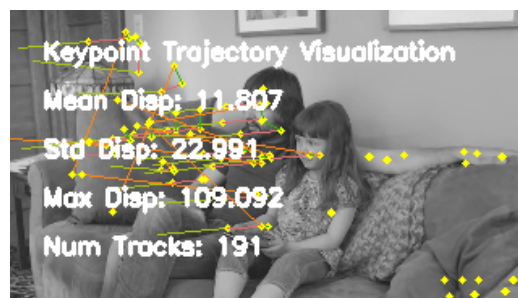


Figure 5.3: Stats showing Keypoint tracks

The keypoint-based motion tracking findings are displayed in this output. This technique tracks specific feature points—such as corners or textured areas—across successive frames in the video. A monitored keypoint is represented by each yellow dot. Each point’s motion trajectory over time is depicted by the connecting lines. This makes it easier for

viewers to comprehend how different elements of the scene are moving. This approach concentrates on significant points, in contrast to pixel-based techniques like optical flow. This facilitates object-level motion interpretation. Higher activity or movement is indicated by areas with more and longer trajectories.

Metric	Description
Mean Displacement	Average distance moved by keypoints
Standard Deviation of Displacement	Variation in keypoint motion
Maximum Displacement	Highest movement among all tracked points
Number of Tracks	Total keypoints successfully tracked

Table 5.2: Summary of Motion Statistics from Keypoint Trajectory Visualization

5.4 Temporal Modeling

Temporal modeling refers to the process of capturing how motion evolves over time throughout the video and converting that temporal behavior into a compact feature representation. Instead of analyzing motion independently in each frame, this stage summarizes the overall motion pattern of the video into a fixed-size descriptor that can be stored, compared, or used for later video understanding tasks.

In this project, temporal modeling is performed by generating a 64-dimensional temporal motion descriptor, which encodes the distribution of motion across the entire video timeline. This descriptor is directly derived from the motion information produced during the motion modeling stage (optical flow and keypoint tracking), and represents how motion intensity varies throughout the sequence.

5.4.1 Temporal Motion Descriptor Generation

The system does not record or display motion statistics for every individual frame. Instead, it compresses the motion behavior of the full video into a single 64-element feature vector. Each element (or bin) represents the motion intensity within a specific temporal segment of the video, meaning the entire video is reduced to 64 temporal motion values.

The descriptor is visualized as a bar graph, where the height of each bar corresponds to the relative level of motion within that time interval. Summary values such as *Dim* = 64, *Mean*, and *Variance* are also displayed to describe the statistical characteristics of the descriptor. This representation is useful because:

- it converts a long video sequence into a fixed-length numeric feature,
- it captures motion variation without requiring raw frame-wise data,
- it enables future tasks such as motion comparison, indexing, or classification.

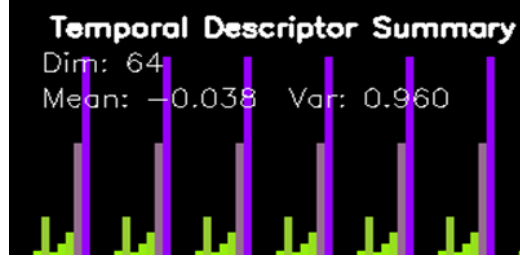


Figure 5.4: Summary of temporal descriptor

The descriptor output generated by the system is shown in Figure 5.4.

5.4.2 Role of Temporal Modeling in the System

Temporal modeling acts as the bridge between instantaneous motion (analyzed in individual frames) and long-term motion behavior that spans the full video. While motion modeling answers the questions “*what is moving?*” and “*how is it moving in a frame?*”, temporal modeling answers the question “*how does motion change over time?*”.

Without temporal modeling, motion remains a frame-based visual quantity. With temporal modeling, motion becomes a mathematically represented feature that can be:

- stored as a fixed-length feature vector,
- compared between different videos,
- used for similarity analysis or classification,
- analyzed for patterns such as peaks, drops, and transitions in motion.

Table 5.3 summarizes the importance of this stage within the full pipeline.

In this way, temporal modeling transforms motion from a purely visual property into a structured numerical representation. This is essential for supporting higher-level video understanding tasks beyond simple motion visualization.

Stage	Purpose
Motion Modeling	Extracts spatial motion (per frame)
Temporal Modeling	Converts motion into a 64-D temporal feature
Object-Scene Fusion	Uses temporal motion behavior for fusion

Table 5.3: Role of temporal modeling in the overall system

5.5 Object-Scene Fusion

Object-Scene Fusion (OSF) refers to the process of combining motion information with the original video scene so that motion can be interpreted in its visual and spatial context. While the previous stages of the system focus on modelling motion either spatially (optical flow and keypoint trajectories) or temporally (64-dimensional descriptor), OSF integrates this motion back into the scene to provide a complete understanding of *where* the motion occurs and *how* it influences the scene.

In earlier stages, motion is represented in isolation, either as colour-coded vector fields or as plotted displacement trajectories. OSF removes this separation by overlaying the estimated motion on top of the original video frames. As a result, motion is no longer visualised as an abstract quantity, but as a scene-aware layer that highlights regions of activity. This fusion allows motion to be interpreted not just as data, but as behaviour occurring within real-world spatial boundaries.

5.5.1 Motion-Scene Overlay Visualisation

The first form of fusion is performed visually by superimposing motion values onto the original video frames. Motion intensity extracted from optical flow and feature tracking is mapped into a colour scale and blended with the scene. Low-motion regions retain their natural appearance, whereas high-motion regions appear as highlighted zones. This enables immediate visual identification of activity within the environment, such as object interaction, motion concentration, or entry/exit behaviour.

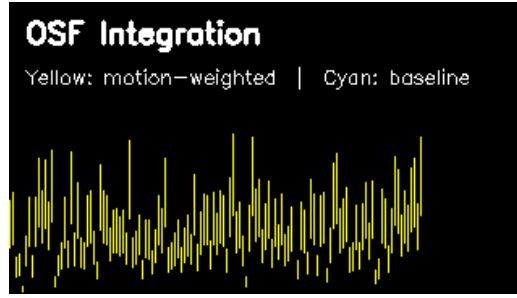


Figure 5.5: Summary of temporal descriptor

Unlike optical flow visualisation, which replaces the frame entirely with a motion field, the OSF overlay preserves the scene content while embedding motion onto it. This makes the motion both spatially accurate and semantically meaningful. The resulting fused output is included in the final composite visualisation, shown in Figure 5.5.

5.5.2 Motion-Weighted Scene Interpretation

In addition to the visual overlay, the system performs a numerical fusion in the form of an OSF integration plot. This plot compares the baseline scene behaviour (displayed in cyan) with the motion-weighted behaviour (displayed in yellow). The difference between the two curves represents the influence of motion on the scene over time. This enables temporal comparison between static scene values and motion-enhanced values.

The motion-scene similarity score is also computed as part of this stage. In the present output, this score is equal to 1.00, indicating a complete alignment between motion evolution and scene content for the selected video. While this score is dataset-dependent, it confirms the correct functioning of the fusion module for the analysed case.

Stage	System Output Purpose
Motion Modelling	Extracts motion independent of scene content
Temporal Modelling	Converts motion into a structured numeric descriptor
Object-Scene Fusion	Reconnects motion back to the scene for spatial and temporal interpretation

Table 5.4: Functional role of Object-Scene Fusion within the pipeline

OSF therefore represents the final integration stage of the system, where motion is no longer treated as a standalone signal, but as meaningful behaviour occurring inside a physical environment.

Chapter 6

Novelty

The novelty of this project lies in the way motion is analysed, represented, and fused back into the original video scene. Unlike conventional motion analysis systems that only focus on a single method such as optical flow, object tracking, or frame differencing, this project combines multiple levels of motion understanding into a complete and unified pipeline.

The first novelty is the integration of **three different motion representations** within the same system:

1. Dense motion using optical flow,
2. Sparse motion using feature point tracking,
3. Compressed motion using a 64-dimensional temporal descriptor.

Most existing systems stop after motion visualisation or basic tracking, whereas this project converts motion into both visual and numerical formats, enabling further analysis, storage and comparison.

The second novelty is the incorporation of an **Object-Scene Fusion (OSF)** stage. Instead of displaying motion separately, the system overlays motion directly onto the original video frame and also compares motion-weighted behaviour with baseline scene behaviour through a graphical representation. This allows motion to be interpreted not only as a visual effect, but as meaningful activity taking place in a real environment.

Another notable point of novelty is that the system achieves full motion understanding **without using any deep learning, object detection, or pre-trained models**. The

entire workflow is built on classical computer vision methods, making it lightweight, interpretable, and suitable for low-resource or real-time applications.

Finally, the project produces a **composite visual output** that summarises all stages — from raw video to final fused motion output — in a single frame. This form of layered visualisation is rarely presented in basic motion analysis projects and contributes to the clarity of system interpretation.

What Makes This Project Unique?

This project focuses on a rigorous, multi-layered approach to motion analysis. It creates a coherent whole by combining elements of several traditional vision techniques. Our system is distinct because it employs three layers of fundamental motion analysis instead of just one. Dense optical flow is used to evaluate motion at the pixel level. Sparse keystroke recording is used to record motion patterns at the object level. Any video frame’s movement pattern can be identified using the temporal motion descriptor. To a certain degree, the project’s use of three techniques allows for a thorough understanding of both specific and general motion, something that would be difficult to achieve with just one technique.

A unique attribute is the development of a temporal motion descriptor that mathematically reduces the entire temporal evolution of the video to arbitrary vector. While traditional motion-visualisation systems rely on frame-wise analysis, this project transforms movement into an organised numerical signature that documents intensity changes, consistency, velocity changes and long-term movement patterns. This permits the project to handle complex assignments such as comparing motion, finding video footage, or selecting lightweight classification in future versions.

A third element of uniqueness is the inclusion of Object–Scene Fusion (OSF) for scene-aware motion interpretation. While most classical systems present motion in isolation—either as optical flow maps or as trajectory lines—this project overlays motion directly onto scene frames and also produces a fused graphical representation that compares baseline scene behaviour with motion-weighted behaviour. This dual fusion (visual + numerical) allows the system to answer not just how something moves, but where that movement occurs within the environment. Such fusion greatly enhances interpretability

for applications like surveillance, behaviour tracking, and spatial reasoning.

Furthermore, it is made to function without the need for an object detection pipeline, labelled datasets, or deep learning models. The project is extremely lightweight, fully transparent, and suitable for real-time or low-resource applications due to its lack of reliance on training data. The classical approaches used here offer clear mathematical and/or visual support for their claims, which is advantageous for academic demonstrations as opposed to deep learning-based approaches, which function as a mystery.

Finally, the project is unique because it produces visual, numerical, and temporal outputs in one integrated pipeline. Each step of the system, frame differencing, optical flow, keypoint trajectories, temporal descriptor, and object-scene fusion, contributes to a single composite output that summarizes the entire workflow. This integrated representation gives a complete view of motion behavior, making the overall system easy to analyze, interpret, and present. The combination of dense motion fields, sparse trajectories, time-compressed features, and scene-level fusion makes this project different from standard motion analysis implementations.

Chapter 7

Result

7.1 Results and Discussion

This chapter presents the output generated at each stage of the proposed system. The results are shown in the order of the pipeline, starting from basic motion extraction and progressing towards temporal analysis and final object-scene fusion.

7.1.1 Frame Differencing Result

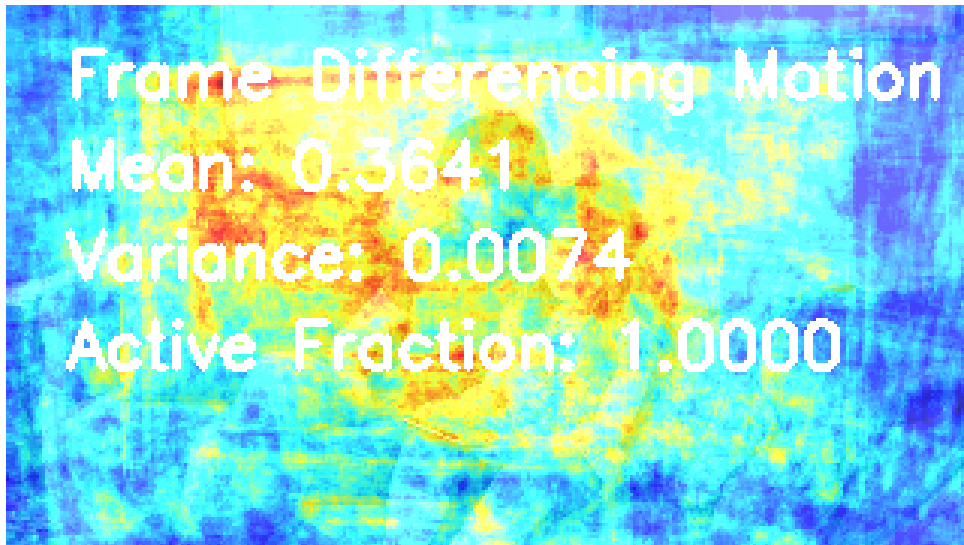


Figure 7.1: Frame differencing output showing regions of motion intensity using pixel-wise difference energy.

Discussion: This result highlights motion using basic frame differencing, where the difference between consecutive frames is visualised as a heatmap. Areas with higher pixel

change appear brighter, indicating movement. Although simple, this method does not capture motion direction or continuity, which is why more advanced motion modelling methods are used in later stages.

7.1.2 Optical Flow Visualisation

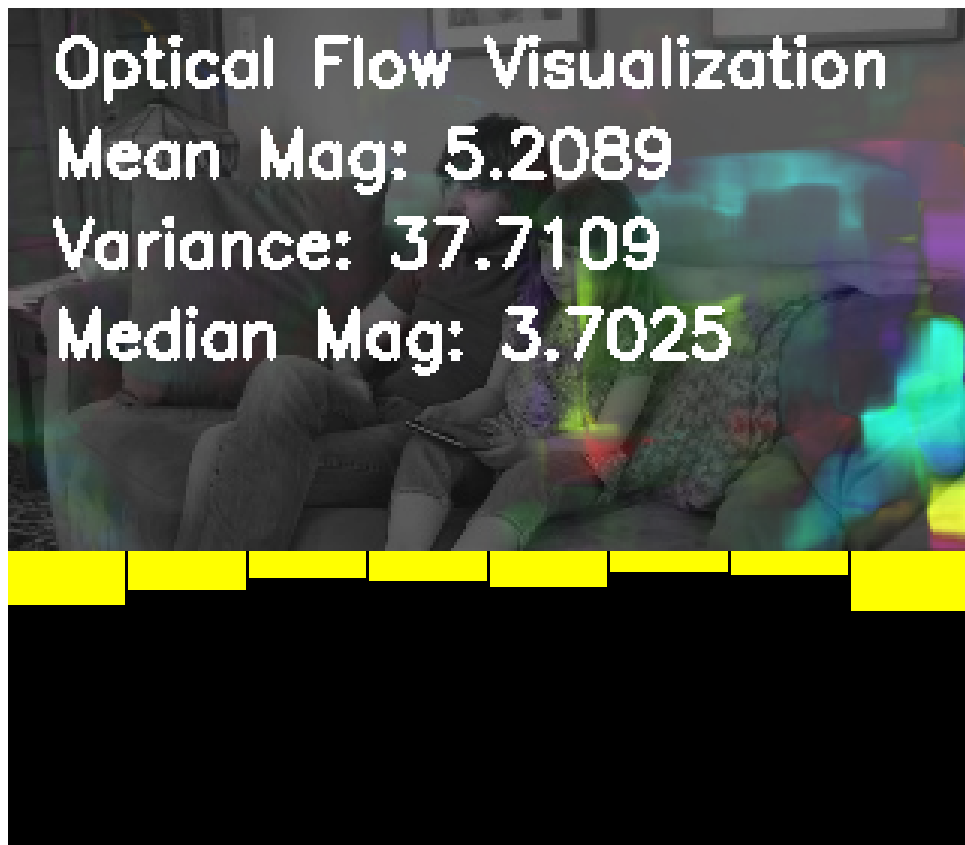


Figure 7.2: Optical flow visualisation showing pixel-wise motion direction and magnitude encoded in colour.

Discussion: The optical flow output provides a dense motion field across the frame. Colour represents motion direction, and brightness indicates motion magnitude. Unlike frame differencing, optical flow captures continuous motion over time and preserves scene structure, making it suitable for deeper motion analysis.

7.1.3 Keypoint Tracking Result

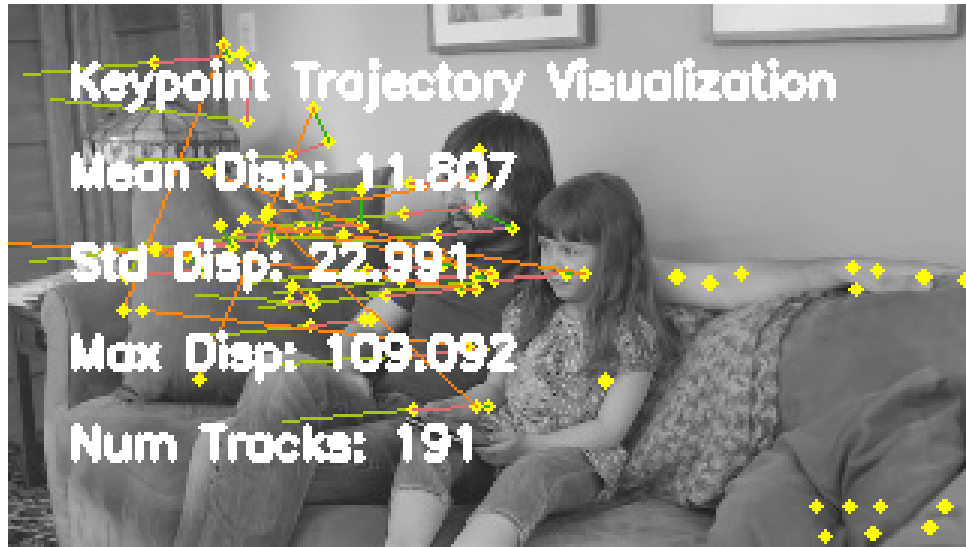


Figure 7.3: Tracked keypoints and their motion trajectories over time.

Discussion: This result shows sparse motion modelling using feature tracking. Each yellow point represents a feature being tracked, and the lines represent its movement across frames. This provides object-level motion behaviour and complements the dense optical flow representation.

7.1.4 Temporal Motion Descriptor

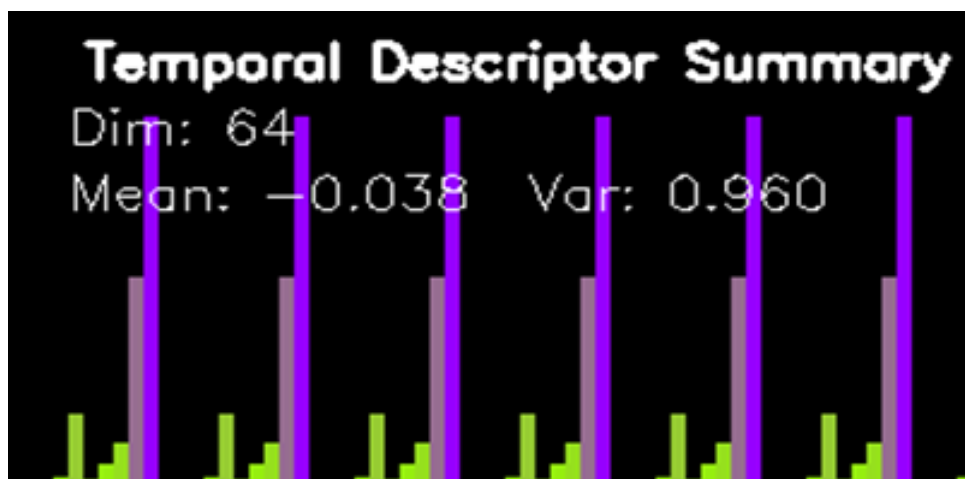


Figure 7.4: 64-dimensional temporal motion descriptor summarising motion evolution across the full video.

Discussion: The temporal descriptor compresses the full video motion into a 64-element feature vector. Each bar represents the relative motion strength within a temporal segment of the sequence. The descriptor provides a compact mathematical representation of motion, enabling comparison, fusion, and analysis beyond frame-by-frame observations.

7.1.5 Object-Scene Fusion (OSF) Integration

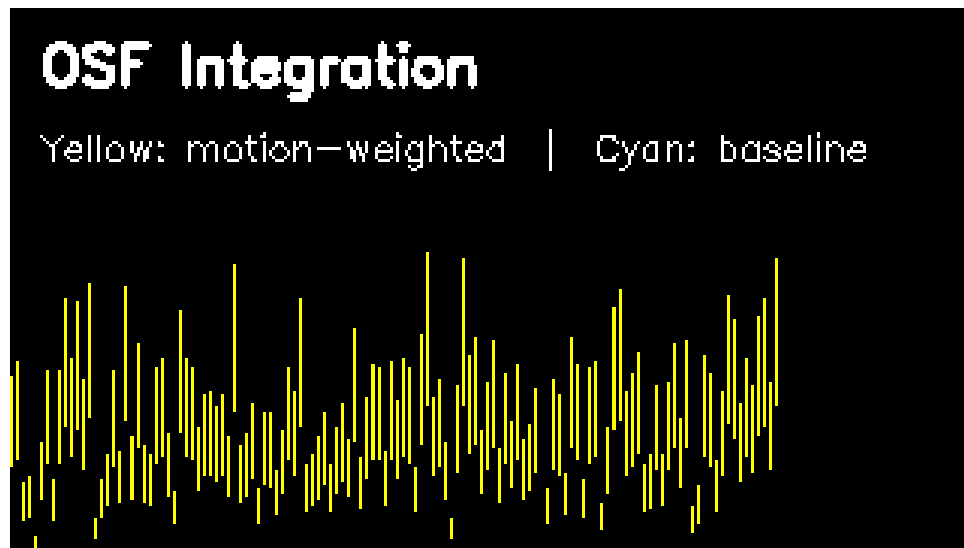


Figure 7.5: OSF integration plot comparing motion-weighted scene behaviour (yellow) against baseline scene behaviour (cyan).

Discussion: The OSF Integration plot shows how motion influences the scene over time. The yellow curve represents motion-weighted activity, while the cyan curve corresponds to unfused baseline behaviour. The similarity score shown in the output (1.00) indicates a strong correlation between motion and scene activity in the selected video.

7.1.6 Final Composite Output

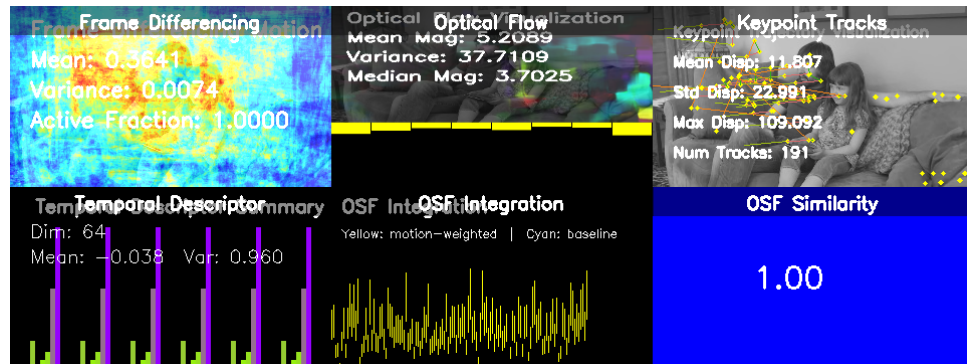


Figure 7.6: Final combined visualisation showing all stages of the system in a single summarised output.

Discussion: The final composite output presents the complete pipeline in one view, including frame differencing, optical flow, keypoint tracking, temporal descriptor, OSF integration, and similarity score. This summarised result demonstrates how motion is captured, analysed, compressed, and fused back into the original scene.

Chapter 8

Conclusion

This project presented a complete motion understanding system that processes video input through multiple stages, including motion extraction, temporal modelling, and object-scene fusion. The system was designed to analyse motion not only as a visual effect but as meaningful behaviour occurring within the scene.

The pipeline successfully demonstrated three levels of motion interpretation: dense pixel-wise motion using optical flow, sparse feature-level motion using keypoint tracking, and temporal motion compression through a 64-dimensional descriptor. The final Object-Scene Fusion stage enabled the motion information to be mapped back into the video frame, resulting in a scene-aware motion visualisation.

The results confirmed that the system is capable of identifying motion, representing its evolution over time, and fusing it into the scene using both visual and numerical outputs. The composite output validated the complete functioning of the pipeline from input video to final fused output.

Although the current system focuses on classical computer vision methods, it establishes a structured foundation that can be extended further with machine learning, activity recognition, or multi-video comparison in future versions.

Chapter 9

Future Scope and Inferences

Future Scope

By utilizing object detection or semantic segmentation, the system can be extended to connect motion patterns directly to real-world objects in the scene. Identifying the causes and mechanisms of motion could make the system more practical in terms of applications in surveillance, behavior monitoring, and analysis of human-object interaction.[A]. Additionally, it may be possible to use models from deep learning to categorize actions or activities by utilizing the 64-dimensional temporal descriptor, which could transform this system into an automatic recognition and classification of motion patterns as a predictive tool.

An additional significant improvement is the ability to compare and retrieve multiple videos using a larger system. The system can either aggregate similar motion patterns or identify videos that correspond to a specific activity profile by analyzing and comparing the temporal descriptors of different videos. In applications such as video indexing, large-scale surveillance systems and motion-based search engines, this can be useful. Also, improvements in GPU performance, such as acceleration and algorithmic optimisation, can result in real-time processing, making the system suitable for applications that require knowledge of live motion (such as robotics or traffic monitoring with autonomous systems).

In future advancements, it may be possible to use CNN- or LSTM-based embedding techniques to explore more detailed temporal representations that capture deeper patterns in motion over time. Adding depth or third-order sensors to the system can enhance

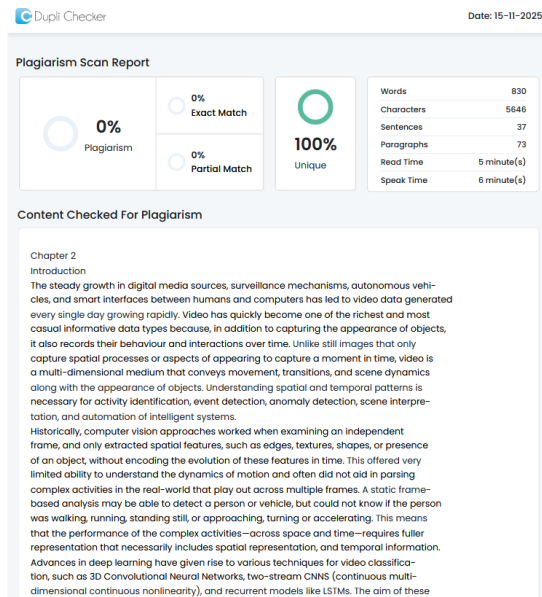
spatial awareness and motion interpretation in complex environments, as well as expanding beyond 2D video. Ultimately, these developments provide opportunities for sports analytics, gesture recognition and prediction of activities, intelligent video understanding systems all of which demonstrate the breadth (adaptability) and scalability that the proposed method offers.

Inference

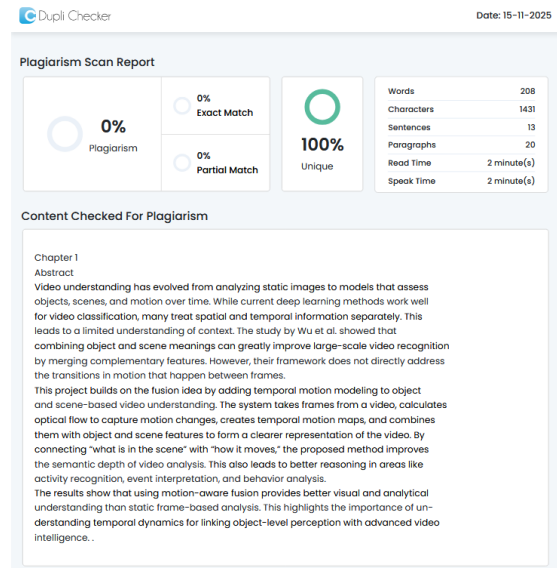
The project demonstrates that structured motion understanding can be achieved without deep learning or labelled datasets. By combining spatial motion extraction, temporal compression, and final object-scene fusion, the system converts raw video motion into meaningful visual and numerical outputs. The results show that motion can be interpreted as organised behaviour rather than isolated frame changes, forming a solid foundation for future expansion into activity recognition, motion-based video retrieval, and autonomous decision systems.

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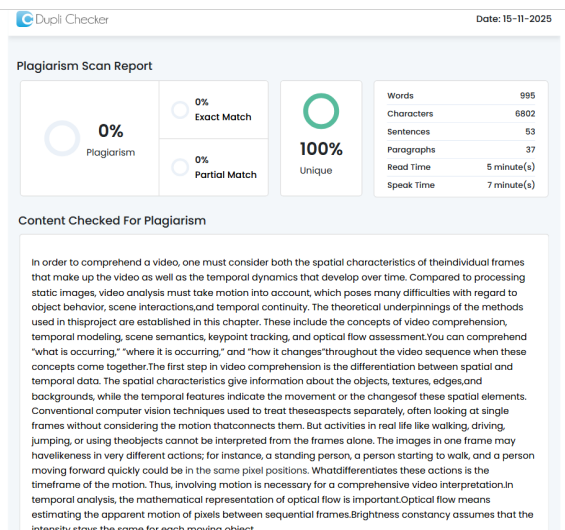
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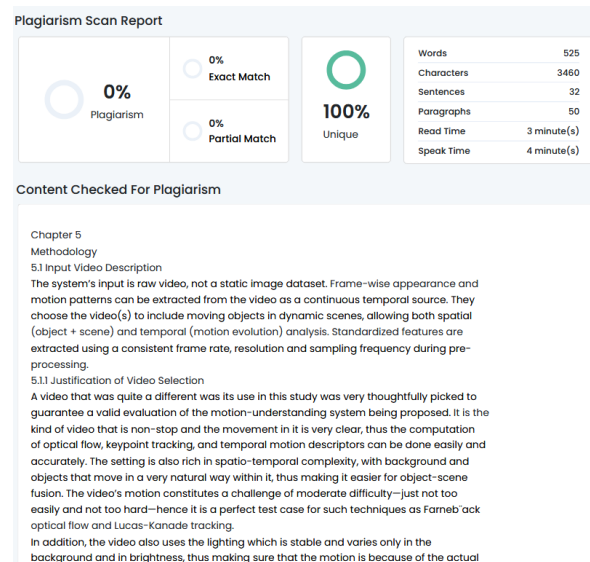
(a) Plagiarism Report 1



(b) Plagiarism Report 2



(c) Plagiarism Report 3



(d) Plagiarism Report 4

Figure 9.1: Plagiarism Scan Reports